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BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Ethical Analysis Report

Group ID: R26-DS-011

Project Title: AI Based Emergency Call and SMS Triage System for Disaster Response in Sri Lankan Low Resource Local Languages

	Identification and Analysis
Data Privacy Concerns	**Issue/Risk:** The system handles some of the most sensitive data a person can share: voice recordings and text messages sent during an emergency. These contain voices (biometric data), phone numbers, home addresses, names of family members, injuries, and the fact that a household is trapped or in danger. The Disaster Management Centre (DMC) provided 60 real emergency call recordings from the November-December 2025 floods and landslides. One recording was corrupted and could not be processed, so it was excluded, leaving 59 usable calls. These are real people speaking in distress, so the privacy risk is far higher than for simulated data. If leaked, a recording could identify a caller by voice or location. Live operation adds further personal data: SMS sender numbers (Twilio), caller audio saved per turn, and fused incident records that link several reports to one household. **Mitigation/Control:** (1) DMC data stays inside the research group: the recordings are used only with the DMC's permission and only for this academic study. They are not shared with anyone outside the research group, are stored locally, and are never uploaded to GitHub, cloud drives or third-party services; the audio/ folder is excluded through .gitignore. (2) Pseudonymisation: each DMC call is stored under its numeric recording ID only (e.g. 890001000492658.wav). The metadata file keeps only what the research needs (urgency label, disaster type, language, district, recording quality), with no caller name, phone number or exact address. (3) Data minimisation: the DMC calls are used only to test the urgency model; the corrupted recording was excluded and not used in any form. They were never used to train any model, and speaker diarization keeps only the caller's turns for analysis. (4) Publications report only aggregate results (accuracy, F1) and short anonymised transcript fragments; no audio is published. (5) The development corpus (267 calls, 864 SMS, 535 fusion reports) is simulated or synthetic, so model development did not depend on real victims' data. (6) Retention: the DMC recordings and all derived files (cleaned audio, caller-audio segments, transcripts) will be deleted once they have been used for the evaluation.
Data Security	**Issue/Risk:**

	The integrated system exposes several services: the call agent (port 5010), the SMS service (5020/5021), the urgency web demo (5000) and the incident fusion API (8000) with its operator dashboard. They exchange personal emergency data over HTTP. The project also relies on external APIs (OpenAI, Google Places, Twilio) whose keys are stored in .env files. A leaked key, an unsecured endpoint or a stolen laptop could expose real emergency reports, allow false reports to be injected, or let an attacker read the incident map of where vulnerable people are. **Mitigation/Control:** (1) Source code is kept in private Git repositories. API keys live only in git-ignored .env files, each with a key-free .env.example template. Raw DMC audio, trained model weights and live report stores (live_reports.jsonl, incident stores) are also git-ignored. (2) Component 4 accepts reports only in the versioned c4.v2 contract; unexpected fields are rejected (HTTP 422), which blocks malformed or tampered input. (3) In the prototype all services run on localhost on the research laptop; only the Twilio webhooks are exposed, through a temporary tunnel used during testing. (4) Trained artifacts are verified with SHA-256 hashes and PROVENANCE files before installation, so a modified model is detected. (5) Before any real deployment: HTTPS, authenticated and role-based access to the dashboard and APIs, encrypted storage for recordings and incident records, audit logs, and hosting on a DMC-controlled server rather than a personal machine. Sending report text to external APIs (OpenAI, Google Places) can be turned off (C4_LLM_ENABLED=0, cache-only Places); the mT5 student model was built so that extraction can run without any external API.
Bias and Fairness	**Issue/Risk:** (1) Language bias: the system serves Sinhala, Tamil, English and code-mixed Singlish / Tanglish speakers, but performance is not equal. Sinhala speech recognition is weaker on real telephony audio, the NER model scores differently per language, and Google Places locates Sinhala / Tamil script places less reliably (53-58% within 1 km vs 83% for Latin script). (2) Real-data imbalance: 58 of the 59 evaluated DMC calls are in Sinhala and only one is in Tamil, so performance for Tamil-speaking callers is largely untested on real calls. (3) Scenario and region bias: the simulated training calls focus on fires and floods, and the DMC calls come from a limited set of districts (mostly Kandy, Puttalam, Colombo, Gampaha). The urgency model reached 80% accuracy on fire/flood calls but 44.8% on all real calls. (4) Speaker bias: acoustic urgency features (pitch, energy, speaking rate) vary with gender, age and how calm a person naturally is, so a quietly spoken caller in real danger

	could be under-triaged. (5) Keyword-based priority rules (MPDS) originally existed only in English. **Mitigation/Control:** (1) Results are reported per language and per class (not just overall accuracy), and the gap on real DMC calls is stated openly rather than hidden. (2) Class imbalance is handled with SMOTE and class-balanced weights, fitted inside training folds only. (3) The urgency model was frozen before the DMC evaluation, so the real-data results are an honest, untuned measurement. (4) Under-triage is treated as the dangerous error: High-class recall is tracked, the call-level urgency keeps the peak severity, and the SMS LLM safety net can only raise priority, never lower it (dangerous under-triage cut from 11 to 1 of 26 cases). (5) MPDS keywords were translated into Sinhala, Tamil, Singlish and Tanglish. Question templates were reviewed by a native speaker, and native review of the keyword lists is planned. (6) The fusion component never uses urgency or priority to decide whether reports are one incident (enforced by a test), so a report is not merged or split because of who sent it. (7) Future work: more Tamil and non-fire/flood real calls, and wider district coverage.
Transparency and Accountability	**Issue/Risk:** The system makes decisions that affect who gets help first: which question to ask, how urgent a call is, what MPDS priority an SMS gets, and whether two reports are the same incident. If an operator cannot see why the system decided something, a wrong decision could go unnoticed, for example a missed high-urgency call or two separate emergencies merged into one, which could leave a household without help. LLMs used in parts of the pipeline can also invent information (hallucination). **Mitigation/Control:** (1) Human in the loop: the system supports the call-taker and the dispatcher, but does not replace them. Every final dispatch decision is made by a human operator, and uncertain incident links are flagged review_required instead of being merged automatically. (2) Explainable outputs: the urgency result shows class probabilities and confidence; every MPDS code lists the evidence that triggered it; every incident merge shows an evidence trail (location, time, similarity and corroborating or contradicting caller words) on the operator dashboard; SHAP explains the urgency features. (3) Hallucination guards: every LLM-extracted field must quote the original message word for word, and the quote is checked before the value is accepted. (4) Traceability: every model is versioned with run IDs, SHA-256 hashes and provenance files; dataset versions are never edited in place; decision rules and thresholds were written down before results were seen. (5) Honest reporting: negative results and

	limitations (e.g. ASR failure on real telephony audio, the 28% false-split rate in fusion) are documented in the project reports.
Impact on Stakeholders	**Identification of affected stakeholders:** (1) Disaster victims and callers - people reporting emergencies by call or SMS, often injured, trapped or panicking, and speaking Sinhala, Tamil, English or a mix. (2) The Disaster Management Centre (DMC) and its call-takers / dispatchers, who provided the 60 real call records (59 usable after one corrupted recording was excluded) and are the intended users of the system. (3) The individuals whose voices are in the 60 DMC recordings, who called during the 2025 floods and landslides. (4) First responders (fire, police, ambulance, disaster relief teams) who act on the triaged and fused incident records. (5) Tamil-speaking and rural communities, who may be under-served if the system works less well for them. (6) SLIIT supervisors and the academic community, who must be able to reproduce and audit the work. **Impact:** (1) Positive: the system can understand callers in their own language, ask fewer and better follow-up questions, flag the most urgent calls first, work over SMS when voice networks are congested, and merge duplicate reports so the same incident is not dispatched twice while another is missed. This can shorten response times during large disasters. (2) Operator autonomy is preserved: operators see the reasoning and can override every suggestion. (3) Risk: an under-triaged call or a wrong merge could delay help. This is reduced by favouring over-triage, keeping the peak severity, preferring a false split over a false merge (merge precision 98.1%), and sending uncertain cases to human review. (4) Privacy of the DMC callers is protected by pseudonymisation, local-only storage within the research group and deletion after use, and the DMC data is used only for evaluation. (5) Victims who are panicking or tired are protected by distress detection and fatigue mode (simple numbered choices instead of long questions). (6) Equity: known language and region gaps are reported openly so DMC can judge where the system is ready and where human handling must remain the default.